

Instructions:

1. This question paper contains 35 questions. All questions are compulsory.
2. The duration of the examination is 60 minutes.
3. Each question carries 2 marks. There are no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

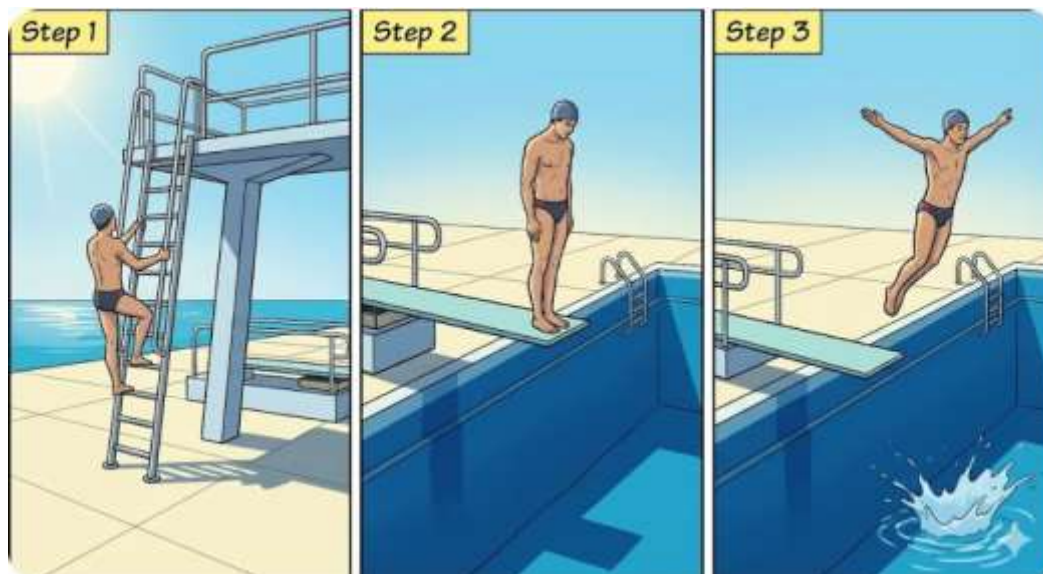
Science

1. Venkat kicks a football on a grassy ground. After rolling for some distance, the ball slows down and eventually stops, even though no one touched it. **Which specific force is responsible for opposing the motion of the ball?**



- (A) Gravitational Force (B) Magnetic Force
(C) Frictional Force (D) Muscular Force
2. In a deep-sea diving mission, divers carry tanks containing a mixture of gases. If they were to use pure oxygen at high pressure, it could be harmful. Instead, they use a mixture where oxygen is diluted, similar to how it is diluted in our atmosphere by **which unreactive gas?**
- (A) Carbon monoxide (B) Nitrogen
(C) Hydrogen (D) Chlorine
3. Three children were diagnosed with different conditions:
Rahul: Cannot see clearly in the dark.
Sita: Has weak bones that bend easily.
Arjun: Has bleeding gums and wounds take time to heal.
Identify the vitamins they are lacking respectively:
- (A) Rahul-Vitamin A, Sita- Vitamin D, Arjun- Vitamin C
(B) Rahul- Vitamin B, Sita- Vitamin C, Arjun- Vitamin D
(C) Rahul- Vitamin C, Sita- Vitamin A, Arjun- Vitamin K
(D) Rahul- Vitamin A, Sita- Vitamin B, Arjun- Vitamin C

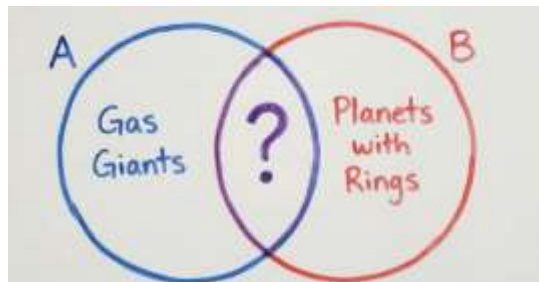
4. During a laboratory experiment, a scientist observes a substance that can be squeezed into a much smaller volume using a piston. Based on its properties, **what is the most likely state of this substance?**
- (A) Solid (B) Liquid (C) Gaseous (D) Water
5. A diver is standing on a high diving board, as shown in the image below. **Observe the situation carefully in three steps.**



At Step 2, what kind of energy does the diver mostly have?

- (A) Kinetic Energy (B) Chemical Energy (C) Potential Energy (D) Heat Energy
6. Seeds of "X" are light and have wing-like structures. Seeds of "Y" are spongy and can float. **Identify X and Y.**
- (A) X = Coconut, Y = Maple (B) X = Maple, Y = Lotus
(C) X = Xanthium, Y = Drumstick (D) X = Pea, Y = Cotton
7. Imagine you have a mixture of oil and water in a beaker. You notice the oil forms a distinct layer on top of the water. To separate these two "immiscible" liquids, **which piece of laboratory equipment would be most effective?**
- (A) A filter paper and funnel (B) A magnetic roller
(C) A separating funnel (D) A sieve with fine mesh
8. Which of the following is the correct difference between a **'Butterfly'** and a **'Cockroach'** regarding their life cycles?
- (A) A butterfly has 3 stages; a cockroach has 4.
(B) A butterfly has 4 stages; a cockroach has 3.
(C) Both have 4 stages, but the cockroach has a pupa stage.
(D) Only the butterfly lays eggs.

9. If Circle A represents "Gas Giants" and Circle B represents **"Planets with Rings"**, **which planet would be placed in the intersection of both circles?**



- (A) mars (B) earth
(C) Saturn (D) Mercury
10. To protect the iron gate of his house from turning into crumbly brown rust, Amit decides to paint it. Painting prevents the iron from reacting with which specific components of the air?
- (A) Nitrogen and Argon (B) Oxygen and Moisture
(C) Carbon Dioxide and Dust (D) Helium and Neon
11. Study the experiment below:
Plant P: Kept in a sealed glass jar with a candle.
Plant Q: Kept in a sealed glass jar without a candle.
 Both were kept in the sunlight. After 24 hours, Plant P grew slightly better than Plant Q. **Why?**
- (A) The candle provided extra heat to Plant P.
 (B) The candle released Carbon Dioxide, which the plant used for faster photosynthesis.
 (C) The candle used up all the Oxygen, leaving more room for the plant to grow.
 (D) The light from the candle was brighter than the sun.
12. In a coastal area, salt is produced by allowing seawater to stand in shallow ponds under the hot sun. As the water disappears into the air, the salt crystals are left behind. **Which process is being used here?**
- (A) Condensation (B) Sublimation
(C) Evaporation (D) Filtration
13. **Assertion:** A person living in a completely dark cave for a year without supplements may develop weak and soft bones.
Reason: The human body needs exposure to sunlight to synthesize Vitamin D, which helps absorb calcium.
- (A) Both A and R are true, and R is the correct explanation of A.
 (B) Both A and R are true, but R is NOT the correct explanation of A.
 (C) A is true, but R is false.
 (D) A is false, but R is true.

14. Match the solar system component in Column I with its unique feature in Column II.

Column I	Column II
P. Saturn	1. The "Red Planet" due to iron rust
Q. Jupiter	2. Has the most visible, beautiful rings
R. Mars	3. Supports life due to liquid water
S. Earth	4. The largest planet with a Great Red Spot

(A) P-2, Q-4, R-1, S-3

(B) P-4, Q-2, R-3, S-1

(C) P-2, Q-1, R-4, S-3

(D) P-3, Q-4, R-1, S-2

15. **Assertion:** If you place a carnation flower with a white stem into a glass of blue-colored water, the petals will eventually turn blue.

Reason: The xylem vessels in the stem carry water and dissolved minerals upward to the leaves and flowers.

(A) Both A and R are true, and R is the correct explanation of A.

(B) Both A and R are true, but R is NOT the correct explanation of A.

(C) A is true, but R is false.

(D) A is false, but R is true.

16. A student observes the Moon in the sky. It looks like a perfect semi-circle. A week later, he sees the full circular disk. **Which phase was he looking at initially?**

(A) New Moon

(B) Full Moon

(C) Crescent Moon

(D) Half Moon

17. Match the chemical process in Column I with its real-world application.

Column I (Process)	Column II (Example)
(a) Filtration	(P) Sea water turning into clouds
(b) Evaporation	(Q) Removing solid impurities from muddy water
(c) Condensation	(R) Melting of glaciers into rivers
(d) Melting	(S) Fog appearing on a car windshield

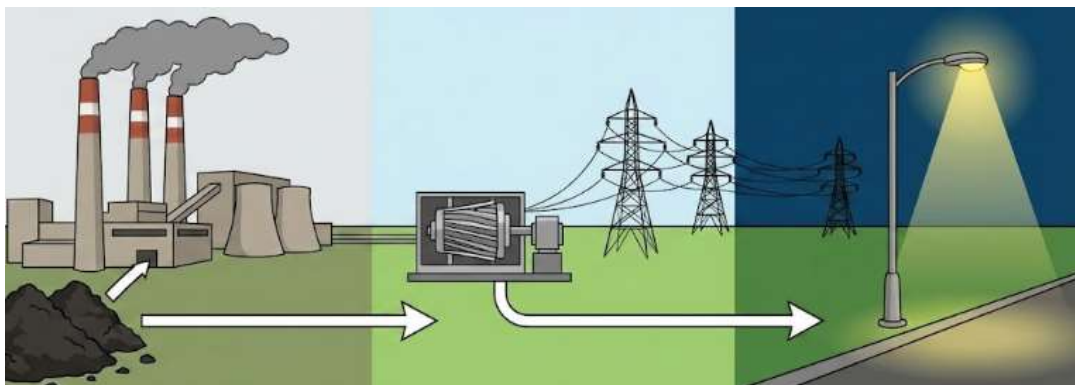
(A) (a)-Q, (b)-P, (c)-S, (d)-R

(B) (a)-P, (b)-Q, (c)-R, (d)-S

(C) (a)-S, (b)-R, (c)-P, (d)-Q

(D) (a)-R, (b)-S, (c)-Q, (d)-P

18. A glass of cold lemonade is placed on a table on a humid afternoon. After a few minutes, tiny droplets of water appear on the outside of the glass. **Where did this water come from?**
- (A) It leaked through the tiny pores of the glass
 - (B) It is the lemonade overflowing from the top
 - (C) Water vapor from the air lost energy upon touching the cold glass
 - (D) The ice cubes inside the glass turned directly into a gas
19. Four seeds (P, Q, R, and S) were tested for dispersal. Seed P has hooks, Seed Q has wings, Seed R is spongy and light, and Seed S is found inside a fleshy fruit. Identify the dispersal agents.
- (A) P-Wind, Q-Animal, R-Water, S-Explosion
 - (B) P-Animal, Q-Wind, R-Water, S-Animal
 - (C) P-Water, Q-Wind, R-Animal, S-Water
 - (D) P-Animal, Q-Water, R-Wind, S-Animal
20. While exploring a volcanic site, an adventurer finds a rock that looks like shiny black glass. He remembers that this rock formed from lava cooling so quickly on the surface that crystals did not have time to grow. **What type of rock did he find?**
- (A) Granite
 - (B) Sandstone
 - (C) Obsidian
 - (D) Marble
21. A girl living in a crowded, poorly ventilated room developed a persistent cough with blood in her sputum and constant fever. This disease is most likely caused by:
- (A) A virus spread through contaminated water.
 - (B) A bacterium spread through the air.
 - (C) A fungus found in the soil.
 - (D) A protozoan spread by a mosquito.
22. A coal-fired power plant supplies electricity to light a street lamp. Which option correctly shows the starting energy source, the intermediate transformation, and the final useful output?



- (A) Solar Energy → Heat Energy → Electrical Energy
- (B) Kinetic Energy → Electrical Energy → Heat Energy
- (C) Solar Energy → Electrical Energy → Light Energy
- (D) Chemical Energy → Electrical Energy → Light Energy

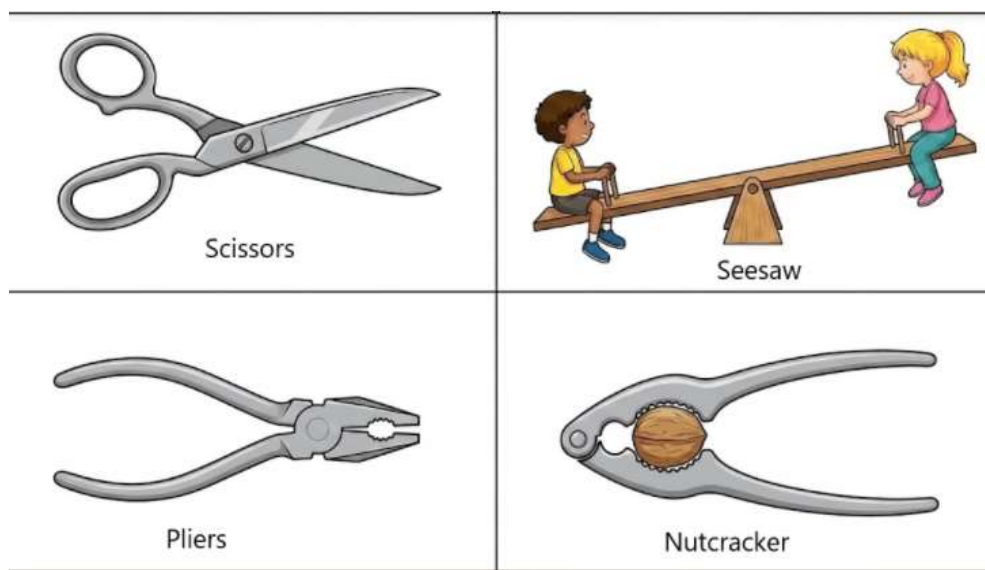
23. A scientist is investigating a "mystery powder" found in a kitchen. When he adds a drop of copper sulphate and caustic soda, the powder stays blue. When he adds iodine, it stays brown. However, when he rubs it on a piece of brown paper and warms it, a permanent greasy spot appears. **The powder is most likely:**

- (A) Powdered sugar
- (B) Groundnut flour (rich in fats)
- (C) Crushed rice (rich in starch)
- (D) Dried egg white (rich in protein)

24. A person's "Small Intestine" is significantly shorter than normal due to a surgery. This person will likely face problems with:

- (A) Chewing food properly
- (B) Absorbing nutrients into the blood.
- (C) Storing undigested waste.
- (D) Breathing deeply.

25. Identify the odd one out based on the class of Simple levers involved:



- (A) Scissors
- (B) Seesaw
- (C) Pliers
- (D) Nutcracker

26. Two friends, Aryan and Kabir, have different health issues. Aryan gets tired easily and looks pale. Kabir has swollen, bleeding gums. **Which of the following represents the correct meal addition for each?**

- (A) Aryan: Citrus fruits; Kabir: Spinach and Liver.
- (B) Aryan: Spinach and Jaggery; Kabir: Amla and Oranges.
- (C) Aryan: Milk and Eggs; Kabir: Rice and Bread.
- (D) Aryan: Iodized salt; Kabir: Sunlight and Butter.

27. A student is testing the solubility and properties of different mixtures. She records her observations in the table below:

Substance added to Water	Observation after Stirring	Visible Parts	Type of Mixture
Salt	Dissolves completely	No	X
Sand	Settles at the bottom	Yes	Y

Based on the data provided, what do X and Y represent?

- (A) X = Heterogeneous, Y = Homogeneous
 (B) X = Homogeneous, Y = Heterogeneous
 (C) X = Solution, Y = Alloy
 (D) X = Solvent, Y = Solute
28. A student learned that different parts of the brain control different activities. During an accident, a person's medulla oblongata was seriously damaged. Soon after, the person needed immediate life support.
 Which problem would most likely occur due to this injury?
 (A) Difficulty in remembering names and events
 (B) Trouble in maintaining balance while walking
 (C) Stoppage of involuntary actions like breathing and heartbeat
 (D) Inability to see objects clearly
29. **Assertion (A):** Burning a piece of wood is considered a chemical change.
Reason (R): The wood transforms into new substances like ash and smoke that cannot be turned back into wood.
 (A) Both assertion and reason are true, and reason is the correct explanation for assertion.
 (B) Both assertion and R are true, but reason is not the correct explanation for assertion.
 (C) Assertion is true, but reason is false.
 (D) Assertion is false, but reason is true.
30. Read the two statements below and choose the correct option.
Statement 1: Gravity only acts on objects that are touching the ground.
Statement 2: If you throw a ball upwards, gravity slows it down until it stops and falls back.
 (A) Both statements are True
 (B) Statement 1 is True, Statement 2 is False
 (C) Statement 1 is False, Statement 2 is True
 (D) Both statements are False

31. Rahul had a common cold, and two days later, his desk mate, Karan, also caught it.
Which of the following "Path" did the germ most likely take?
 (A) It traveled through Karan's drinking water.
 (B) It was transferred through the air when Rahul sneezed.
 (C) It was passed through a mosquito that bit both of them.
 (D) It was caused by Karan not eating enough vitamins that day.
32. Which of the following actions involves Work being done against Gravity?
 (A) Rolling a ball on a flat floor.
 (B) Lifting a bucket of water from a well.
 (C) Pushing a heavy box across a smooth room.
 (D) Sleeping on a bed.
33. The Earth's atmosphere is not just air; it is a protective shield. The Stratosphere layer contains a special form of gas called Ozone. This Ozone acts like a pair of 'sunglasses' for the planet, absorbing harmful radiation from the sun before it can reach the surface where we live.
What would most likely happen to living beings if the "chemical shield" in the Stratosphere was removed?
 (A) The air would become too thin to breathe
 (B) Harmful ultraviolet radiation would cause severe health damage
 (C) The Earth would immediately freeze into solid ice
 (D) All the water on Earth would turn into nitrogen gas
34. The Earth takes approximately $365 \frac{1}{4}$ days to complete one revolution. We count a normal year as 365 days. If a child is born on February 29, 2024, in **which year will they likely celebrate their first real birthday on the actual date?**
 (A) 2025 (B) 2026 (C) 2027 (D) 2028
35. Sarah is making a fruit punch for her birthday party. She stirs sugar into a large pitcher of water until it disappears, and then adds slices of orange and floating ice cubes.
Which part of Sarah's punch represents a solution?
 (A) The floating ice cubes
 (B) The orange slices floating in the water
 (C) The sugar dissolved in the water
 (D) The empty pitcher before adding water

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